*ms*2: A molecular simulation tool for thermodynamic properties, release 3.0

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Abstract

A new version release (3.0) of the molecular simulation tool *ms*2 [Deublein et al., Comput. Phys. Commun. 182 (2011) 2350 and Glass et al., Comput. Phys. Commun. 185 (2014) 3302] is presented. Version 3.0 of *ms*2 features the implementation of two additional ensembles, i.e. microcanonical (NVE) and isoenthalpic–isobaric (NpH), the calculation of Helmholtz energy derivatives in the NVE ensemble, the osmotic pressure and the Maxwell-Stefan diffusion coefficients of quaternary mixtures, thermodynamic integration as a method for calculating the chemical potential, statistics for sampling hydrogen bonds as well as the ability to carry out molecular dynamics runs for an arbitrary number of state points in a single program execution.

Keywords: molecular simulation, molecular dynamics, Monte Carlo

New version program summary

Program Title: *ms*2

Operating system: Unix/Linux

Licensing provisions(please choose one): CC0 1.0/CC By 4.0/MIT/Apache-2.0/BSD 3-clause/BSD 2-clause/GPLv3/CC BY NC 3.0

Programming language: Fortran90

Has the code been vectorized or parallelized: yes (Message Passing Interface (MPI) protocol and OpenMP Scalability is up to 2000 cores)

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Distribution format: tar.gz

Classification: 7.7, 7.9, 12

Catalogue identifiers of previous versions: AEJF_v1_0 and AEJF_v2_0

Supplementary material:

Journal reference of previous versions: S. Deublein et al., Comput. Phys. Commun. 182 (2011) 2350 and C. Glass et al., Comput. Phys. Commun. 185 (2014) 3302

Does the new version supersede the previous version?: yes

Reasons for the new version: introduction of new features

Summary of revisions: two new ensembles (NVE and NpH), new properties (Helmholtz energy derivatives, chemical potential via thermodynamic integration, osmotic pressure, Maxwell-Stefan diffusion coefficients of quaternary mixtures), new functionalities (detection of hydrogen bonding, the ability to carry out molecular dynamics runs for an arbitrary number of state points in a single program execution)

Nature of problem: Calculation of application oriented thermodynamic properties: vapor-liquid equilibria of pure fluids and multi-component mixtures, thermal, caloric and entropic data as well as transport properties

Solution method: Molecular dynamics, Monte Carlo, various ensembles, grand equilibrium method, Green-Kubo formalism, Lustig formalism

Typical running time: varies (typically from a couple of hours up to days) depending on the specific scenario (system size, calculated properties, number of CPU cores used)

Restrictions: typical problems addressed by $ms2$ are solved by simulating systems containing typically 1000 – 5000 molecules that are modelled as rigid bodies

Documentation: documentation is provided with the installation package and is available at http://www.ms-2.de

1. Microcanonical and isobaric-isoenthalpic ensembles

An ensemble is the set of all theoretically possible microscopic configurations on the molecular level under specific macroscopic constrains. The microcanonical (NVE) ensemble is the set of all configurations that fulfill the condition of having the same number of particles N, volume V and energy E, whereas the isobaric-isoenthalpic (NpH) ensemble represents a system at constant number of particles N, pressure p and enthalpy H. Molecular dynamics (MD) mimics the time evolution of the system by numerically solving Newton's equations of motion for all considered molecules. Because of the nature of this solution, the time is discredized and the method yields microscopic configurations at discrete but consecutive time steps. The Monte Carlo (MC) method is the application of statistical mechanics to describe molecular systems. With this approach, microscopic configurations are generated by random numbers that are potentially accepted such that only relevant and physically meaningful configurations are sampled [1]. The generation and acceptance of these configurations is governed by specific probabilities that are ensemble-dependent and known for essentially every relevant ensemble. For MC simulations, the NVE and NpH ensembles were implemented in $ms2$ 3.0 according to Refs. [2, 3]. For MD simulations, the pressure is kept constant using Andersen's barostat [4] in case of the NpH ensemble. Because the solution of Newton's equations of motion is approximate, the total energy of the system $E = K + U$, which consists of a kinetic K (exclusively molecular velocity dependent) and potential U (exclusively molecular position dependent) contribution, is not rigorously conserved in a standard MD run. Therefore, the translational and rotational velocity of every molecule is rescaled such that the total kinetic energy K fulfils $K = E - U$, where the total energy E is specified and the potential energy U is calculated from the current configuration. For an NpH run, the solution is analogous: velocities are rescaled such that the current kinetic energy K fulfils $K = H - U - pV$, where H and p are specified, U and V are dependent on the current microscopic configuration. This extends the ensembles available in $ms2$ to five: NVT, NVE, NpT, NpH and μVT, where μ is the chemical potential.

For verification purposes, Table 1 contains numerical results for a state point located in the dense fluid region of methyl fluoride modeled by a two-center Lennard–Jones potential [5].

2. Helmholtz energy derivatives in the microcanonical ensemble

The calculation of the Helmholtz energy derivatives

$$A^{\mathrm{r}}_{nm} = (1/T)^n \rho^m \frac{\partial^{n+m} f^{\mathrm{r}}(T,\rho)/(RT)}{\partial(1/T)^n \partial\rho^m}, \tag{1}$$

with the molar residual Helmholtz energy f^{r}, the temperature T, the density ρ and the molar gas constant R were introduced in $ms2$ also for the NVE ensemble up to the order $n = 3$ and $m = 2$. It can be shown that the total reduced Helmholtz energy $f/(RT)$ can be additively separated into an ideal $f^{\mathrm{o}}/(RT)$ and a residual contribution $f^{\mathrm{r}}/(RT)$. The ideal contribution by definition corresponds to the value of $f/(RT)$ when no intermolecular interactions are at work [6]. $f^{\mathrm{o}}/(RT)$ consists of an exclusively temperature and an exclusively density dependent part. The latter has a trivial density dependence $\ln(\rho/\rho_{\mathrm{ref}})$, whereas the former is non-trivial. A formalism that allows for the calculation of these derivatives on the fly from a single simulation run per state point was published by Lustig [7, 8] and was already introduced for the NVT ensemble in the previous release. $ms2$ 3.0 now yields these derivatives also for NVE simulations. Here, however, in contrast to NVT simulations, there are two numerical results for each calculated derivative A_{nm} due to the two possible entropy definitions in statistical mechanics [8]. In any case, these two different sets of results must be identical in the thermodynamic limit ($N \longrightarrow \infty$). In practice, they already agree within their estimated statistical uncertainty for a simulation with around a thousand particles.

Table 1: Comparison of the ensembles implemented in $ms2$ for a dense liquid state point of methyl fluoride (R41). Statistical uncertainties were estimated with the method of Flyvbjerg and Petersen [10]

		Temperature K	Density mol/l	Pressure MPa	Potential energy kJ/mol
MD	NVT	220	27	100.2 ± 0.2	-16.63 ± 0.02
	NpT	220	26.994 ± 0.005	100	-16.65 ± 0.04
	NVE	220.09 ± 0.06	27	100.1 ± 0.2	-16.63 ± 0.01
	NpH	219.8 ± 0.2	26.991 ± 0.004	100	-16.64 ± 0.03
MC	NVT	220	27	100.2 ± 0.1	-16.66 ± 0.01
	NpT	220	27.015 ± 0.002	100	-16.66 ± 0.02
	NVE	220.01 ± 0.03	27	100.2 ± 0.1	-16.65 ± 0.01
	NpH	220.11 ± 0.04	27.005 ± 0.002	100	-16.66 ± 0.01

A detailed description of how to convert these derivatives into common thermodynamic properties is provided in the supplementary material of release 2.0 of $ms2$ [9] and in Ref. [6].

3. Thermodynamic integration

The previously described method of Lustig does not allow for the direct sampling of the chemical potential thus entropic properties like A_{00}. Such an effort requires techniques based on free energy calculation, such as particle insertion and/or thermodynamic integration [1]. Widom's particle insertion method [11] is a conceptually straightforward approach to calculate the chemical potential with low computational cost, both for pure substances and mixtures. The total chemical potential μ_i of species i can be separated into an ideal (o) and a residual (r) contribution the same way as the Helmholtz energy is decomposed (cf. section 1): $\mu_i(T,\rho,x_i) = \mu_i^{\rm o}(T) + RT\ln(N_i/(V\rho_{\rm ref})) + \mu_i^r(T,\rho,x_i)$, where N_i is the number of species i, $\rho = N/V$, $x_i = N_i/N$ and $\rho_{\rm ref}$ is an arbitrary reference density. The part $\mu_i - \mu_i^{\rm o}(T)$ is often referred to as the configurational chemical potential $\mu_i^{\rm conf}$. Widom's method requires the frequent insertion of an additional ($i = N+1$) test molecule into the simulation volume at a random position with a random orientation. At constant temperature and constant pressure or volume the potential energy U_i of this test molecule, i.e. the interaction energy with all other "real" N molecules, yields the configurational chemical potential according to

$$\mu_i^{\rm conf} = \mu_i - \mu_i^{\rm o}(T) = -k_{\rm B}T \ln \frac{\langle V \exp(-U_i/k_{\rm B}T)\rangle}{\langle N_i \rangle}, \tag{2}$$

where $k_{\rm B}$ is Boltzmann's constant. The test molecule is removed immediately after the calculation of its potential energy U_i, thus it does not influence the real molecules in the system in any way. In contrast to the usual convention, the brackets <> have a dual meaning here: They stand for either NVT or NpT ensemble average as well as an integral over all possible positions and orientations of the test molecule added to the system. The density of the system has a significant influence on the accuracy of this method. For state points with a high density, test molecules almost always overlap with real molecules, which leads to a potential energy $U_i \to \infty$ and thus to a vanishing contribution to Eq. (2), resulting in poor statistics and often even to complete failure of sampling. Thermodynamic integration is one solution to overcome the limitations of Widom's particle insertion method. The idea behind calculating the chemical potential by thermodynamic integration is to avoid insertion of the test molecule in system A and perform it in system B for which it is not problematic. Since the chemical potential is a state variable, its change $\mu_{A,i} - \mu_{B,i}$ can be calculated using any path between states A and B, which is represented by the parameter λ. It can be shown [1] that the relation between λ and $\mu_{A,i} - \mu_{B,i}$ is

$$\mu_{A,i} - \mu_{B,i} = \int_{B(\lambda_{\min})}^{A(\lambda_{\max})} \left\langle \frac{\partial U_i(\lambda)}{\partial \lambda} \right\rangle d\lambda \tag{3}$$

for particle i. The bracket <> in this equation denotes NVT or NpT ensemble averages and U_i is the potential energy of particle i that must be a part of the system the same way as the other particles are. The only difference between i and all other particles is that its interaction energy $U_i(\lambda)$ is scaled between states $A(\lambda_{\max})$ and $B(\lambda_{\min})$ with λ. As long as $\partial U_i(\lambda)/\partial\lambda$ can be calculated analytically and sampled during simulation, the actual way of scaling $U_i(\lambda)$ with λ can be chosen arbitrarily because μ is a state variable. The integration with respect to λ is carried out numerically. Assuming that $\mu_{B,i}$ is practically zero or at least can be successfully calculated by Widom's particle insertion method at the end point B using Eq. (2), $\mu_{A,i}$ yields the configurational part of the total chemical potential. The non-linear scaling $U_i(\lambda) = \lambda^d U_i$ for $\lambda_{\min} \leq \lambda \leq 1 = lambda_{\max}$ with an adaptive sampling technique [12] was implemented for MC with d and $\lambda_{\min}$ being input parameters. This adaptive technique allows for the sampling of the entire range of $\lambda_{\min} \leq \lambda \leq 1$ in a single MC simulation run with an arbitrary resolution for the numerical integration. In addition to the standard NVT and NpT ensemble MC moves, the simulation includes changes of λ controlled by a proper MC acceptance criterion, ensuring visits at each discrete λ value in the range $\lambda_{\min} \leq \lambda \leq 1$. For MD simulations, changes of λ are also carried out on the fly in a single simulation but without any acceptance criterion. A detailed description of the parameter setup is given in the supplementary material.

4. Quaternary Maxwell-Stefan diffusion coefficients

*ms*2 employs the Green-Kubo formalism based on the net velocity auto-correlation function to obtain $n \times n$ phenomenological coefficients [13]

$$L_{ij} = \frac{1}{3N} \int_0^\infty \mathrm{dt} \left\langle \sum_{k=1}^{N_i} \mathbf{v}_{i,k}(0) \cdot \sum_{l=1}^{N_j} \mathbf{v}_{j,l}(\mathrm{t}) \right\rangle , \tag{4}$$

in a mixture of n components. Here, N is the total number of molecules, N_i is the number of molecules of species i and $\mathbf{v}_{i,k}(t)$ denotes the center of mass velocity vector of the k-th molecule of species i at time t. Note that Eq. (4) corresponds to a reference frame in which the mass-averaged mixture velocity is zero [13].

Starting from the phenomenological coefficients L_{ij}, the elements of a $(n-1)\times(n-1)$ matrix $\mathbf{\Delta}$ can be defined as [13]

$$\Delta_{ij} = (1-x_i)\left(\frac{L_{ij}}{x_j} - \frac{L_{in}}{x_n}\right) - x_i \sum_{k=1\neq i}^{n} \left(\frac{L_{kj}}{x_j} - \frac{L_{kn}}{x_n}\right), \tag{5}$$

so that its inverse matrix $\boldsymbol{B} = \mathbf{\Delta}^{-1}$ is related to the Maxwell-Stefan diffusion coefficients $Đ_{ij}$. In the case of a quaternary mixture, the six Maxwell-Stefan diffusion coefficients are then given by [13]

$$Đ_{14} = \frac{1}{B_{11} + (x_2/x_1)B_{12} + (x_3/x_1)B_{13}}, \tag{6}$$

$$Đ_{24} = \frac{1}{B_{22} + (x_1/x_2)B_{21} + (x_3/x_2)B_{23}}, \tag{7}$$

$$Đ_{34} = \frac{1}{B_{33} + (x_1/x_3)B_{31} + (x_2/x_3)B_{32}}, \tag{8}$$

$$Đ_{12} = \frac{1}{1/Đ_{24} - B_{21}/x_2}, \tag{9}$$

$$Đ_{13} = \frac{1}{1/Đ_{14} - B_{13}/x_1}, \tag{10}$$

$$Đ_{23} = \frac{1}{1/Đ_{24} - B_{23}/x_2}. \tag{11}$$

(12)

Molecular dynamics simulations for Lennard-Jones fluids were performed in order to test the validity of the quaternary Maxwell-Stefan diffusion coefficients calculated with *ms*2. For this purpose, a quaternary Lennard–Jones pseudo mixture was created with different name labels for the molecules divided into four mole fractions (x_1= 0.1, x_2= 0.2, x_3 = 0.3 and x_4 = 0.4 mol mol^{-1}). A system of 800 Argon molecules was simulated in the dense liquid state and the resulting Maxwell-Stefan diffusion coefficients were compared with the corresponding self-diffusion coefficient that has to be identical with the present case.

Fig. 1 shows the development of the calculated values of the six different Maxwell-Stefan diffusion coefficients with the number of time origins. As can be seen, the resulting values of become independent after around 10^4 time origins and then oscillate around their mean value. The higher statistical uncertainties of $Đ_{12}$ and $Đ_{13}$ are due to the small number of type 1 molecules from compared with the amount of molecules of other types.

5. Hydrogen bonding

There is no unambiguous characterization of a hydrogen bond between two molecules [14, 15, 16]. Rather, the hydrogen (H) bond 'is a structural motif and involves at least three atoms' [17]. The International Union of Pure and Applied Chemistry (IUPAC) defines the hydrogen bond as 'an

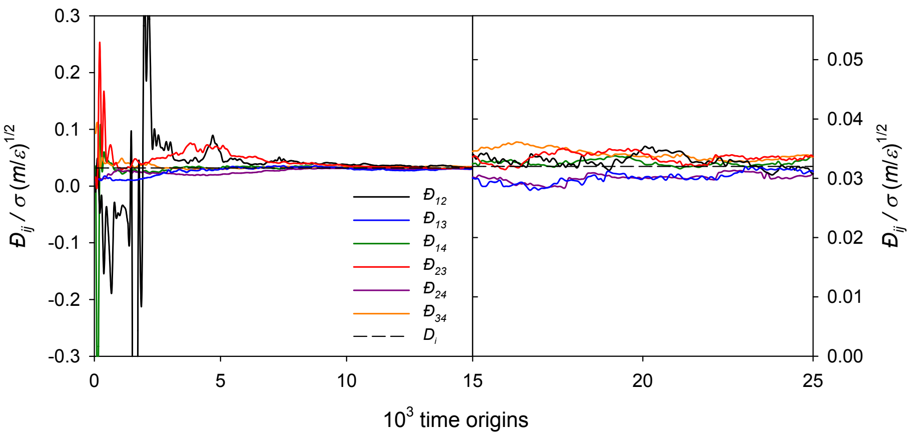

Figure 1: Maxwell-Stefan diffusion coefficients $Ð$ as a function of time origins for a quaternary pseudo mixture ($\epsilon_1 = \epsilon_2 = \epsilon_3 = \epsilon_4$ and $\sigma_1 = \sigma_2 = \sigma_3 = \sigma_4$) at $k_\mathrm{B}T/\epsilon_1 = 0.728$ and $\rho\sigma_1^3 = 0.8442$.

attractive interaction between a hydrogen atom from a molecule or molecular fragment X–H in which X is more electronegative than H, and an atom or a group of atoms in the same or a different molecule, in which there is evidence of bond formation' [16].

Energetic [14], geometric [18] and topological [15] hydrogen bonding criteria have been proposed in the literature. Geometric criteria, which are not overly complex, are often based on the following assumptions [19]:

- The interaction between two hydrogen bonded sites is highly directional and short ranged.
- A donor interacts at most with a single acceptor, an acceptor can interact with multiple donors.

Accordingly, a class of geometric criteria for the evaluation of hydrogen bonding networks in fluids was implemented in *ms*2. Thereby, the triangle between three charge sites, being part of two molecules, is evaluated to determine whether an interaction between two sites constitutes a hydrogen bond or not. A molecule acts as a donor to another molecule, i.e. the acceptor, if the following conditions hold [18, 19, 20, 21], cf. Fig. 2:

- The distance between the donor and the acceptor is smaller than a threshold distance, i.e. ℓ_AD or ℓ_DA.
- The distance between the acceptor sites of the acceptor and donor molecules is smaller than ℓ_AA.
- The angle between the acceptor-donor axis and the acceptor-acceptor axis is smaller than a threshold angle, i.e. φ_DAA or φ_AAD.

Therein, ℓ_{AD}, ℓ_{DA}, ℓ_{AA}, φ_{DAA} and φ_{AAD} are the parameters of the implemented class of hydrogen bonding criteria. For methanol, e.g., Haughney *et al.* [18] proposed $\ell_{\mathrm{AD}} = \ell_{\mathrm{DA}} = 2.6$ Å, $\ell_{\mathrm{AA}} = 3.5$ Åand $\varphi_{\mathrm{AAD}} = \varphi_{\mathrm{DAA}} = 30°$. Such geometric criteria have been applied to a variety of fluids, in particular to water [20, 21, 22], methanol-water mixtures [23, 24] and ethanol [19, 25, 26]. Hydrogen bonds in sorbate-sorbent interactions can be treated analogously [27]. E.g., for hydrogen fluoride, a simpler distance-based criterion can be used [28], which is also covered by the present implementation. A detailed description of the parameter setup is given in the supplementary material.

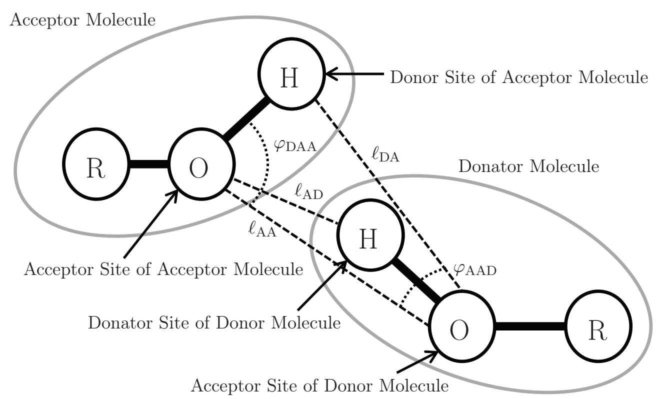

Figure 2: Hydrogen bonding criteria implemented in the present release version of *ms*2.

6. Ewald summation

The applicability of *ms*2 was extended to electrically charged molecules or ions. The long ranged intermolecular interactions between those molecules and the surrounding solvents are considered using two well established approaches, Ewald Summation and smooth-particle mesh Ewald summation. Both approaches are well known [1] and are introduced only briefly here.

In the Ewald summation, the overall coulombic potential acting in the simulation volume is determined by a sum of two terms, the short range coulombic contribution $u^{\mathrm{c,short}}$ and the long range coulombic contribution $u^{\mathrm{c,long}}$ according to

$$u^{\mathrm{c}} = u^{\mathrm{c,short}} + u^{\mathrm{c,long}} \; . \tag{13}$$

The separation into the two terms is achieved by the introduction of a fictive charge density function $\rho^{\mathrm{screen}}(\boldsymbol{r})$, which acts in the entire simulation volume.

Following the Ewald approach, for any configuration of molecules in the simulation volume, each point charge of magnitude q_l in the simulation volume is now superimposed by one countercharge of magnitude $-q_l$. By the presence of this superimposed charge, the interaction potential decays to zero within a distance reasonably low for molecular simulation and can, hence, be explicitly considered, cf. short term contribution of the Ewald summation.

The second term $u^{\mathrm{c,long}}$ in Eq. (13) determines the contributions that were subtracted due to the introduction of the fictive charge density function. This term can not be determined explicitly by the evaluation of pairwise interactions, since the coulombic potential at $r_{lm} = L/2$, the longest distance accessible in molecular simulation, still shows significant contributions to the potential energy in the system. In Ewald summation, these contributions are determined in Fourier space from the negative charge distribution function $-\rho^{\text{screen}}(\boldsymbol{r})$. Since $\rho^{\text{screen}}(\boldsymbol{r})$ depends on molecular motions and hence molecular positions, a Fourier transformation is performed for any change in the simulation volume.

The smooth-particle mesh Ewald summation is widely considered as an improved Ewald summation. In this approach, the concept of splitting the long range charge-charge interactions are fully employed. The difference between both methods is only in the calculation of the long range contributions. In the SPME approach, the electrostatic field of ions in the simulation volume is described by a spline function with a given functional form. For this given spline equation, the Fourier transformation is known and has, hence, not to be determined in each simulation step. This accelerates the calculation of the long range contribution and reduces simulation time and effort. The accuracy of the SPME results is equivalent to the Ewald approach.

7. Parallel ensemble calculations

*ms*2 was already parallelized in its initial version for distributed memory architectures using MPI [29, 9]. The present version adds a new level of parallelization. The calculation of different ensembles is independent of each other. Therefore sampling different state points can easily be done concurrently. To achieve this, the processing elements (PE) are split in c disjunct groups and each group consecutively computes the assigned ensemble. To enable this feature, the *mpiEnsembleGroups* option was introduced for the input file. The default (if not set or set to 0) is the old behavior not to split the PE and to use a single MPI group. A value of 1 will enable the feature and automatically set the number of MPI groups to a minimum of the number of ensembles and the number of PE. Otherwise this option will set the number of MPI groups to the value given. Another change induced applies to the restart capability of *ms*2. A checkpoint now consist of multiple rst-files and can be restarted using the -r (or –restart) option of *ms*2.

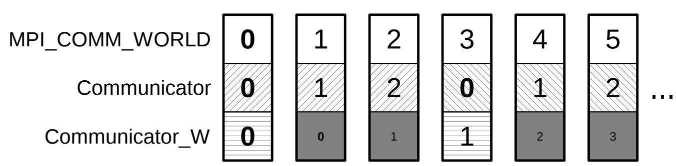

Figure 3: *ms*2 MPI ranks

*ms*2 will create a MPI communicator for each of the aforementioned MPI groups with MPI_Comm_Split (see 3). The "coloring", i.e., assignment of the PE to the MPI communicators is done was continguous blocks. This is advantageous compared to e.g. a round robin fashion if the processes have to be pinned to NUMA domains. Another communicator contains the root processes (subcommunicator rank 0 processes) of all the groups to ease the communication on a higher level among the groups (and there still is MPI_COMM_WORLD).

Even if the computation of different ensembles is embarrassingly parallel, there is still an interaction between the different communicators if e.g. the program gets a signal to terminate properly (to write e.g. a checkpoint before terminating). This signal is received from an arbitrary single PE (or non-isochronic from multiple PE). The previous version, where ensembles were calculated sequentially with a single global communicator, a MPI_Allreduce call took care to spread this information among the processes after each iteration and is still acceptable within the subcommunicators. A collective communication of all PE beyond subcommunicator boundaries for each iteration in the computation implicitly would also synchronize all of the ensemble computations at this point. This is not a satisfactory solution, since computationally less intensive ensembles will be forced to wait for the slower ones. The present implementation uses non-blocking communication between the subcommunicator roots to avoid this.

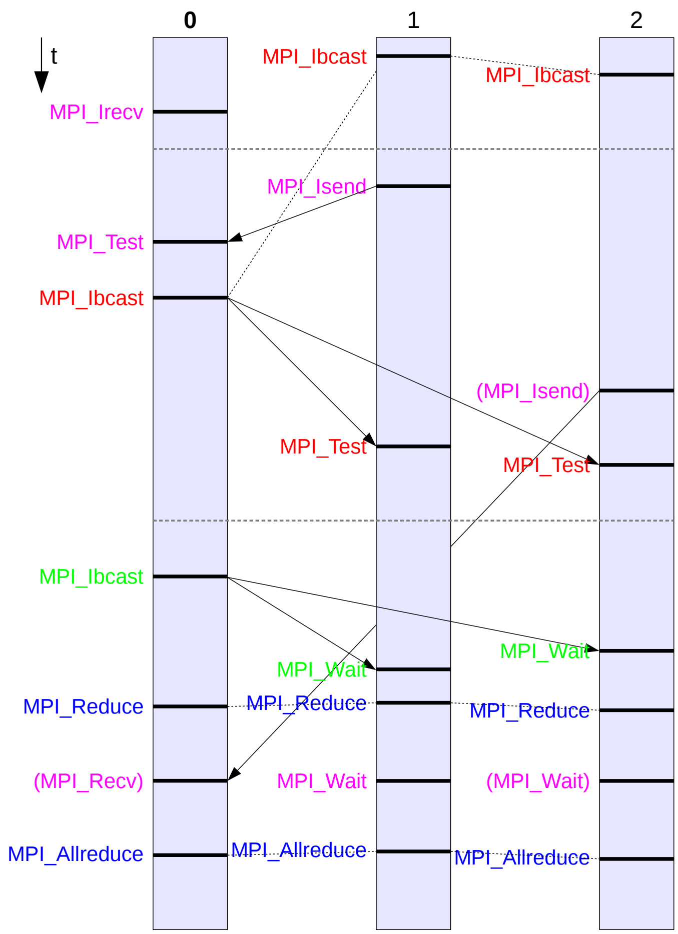

Figure 4: $ms2$ communication to pass termination status to all PEs

Before the iterations start, the root process will execute a MPI_Irecv, whereas all other processes start a MPI_Ibcast (see 4). During the iteration any process might trigger a termination by sending a message to the root, which will broadcast the information to notify all other processes. If no termination happened during the iterations, the root will send a message to itself and broadcast a non-termination message to all other processes to finish the outstanding receive and broadcast. To take care of several processes sending a termination message at once, a summation reduction determines the sum of all these messages for the root process to receive them all. This technique can also be used hierarchically, replacing the MPI_Allreduce within each subcommunicator.

After a termination, every subcommunicator writes its own restart (*.rst) file.

8. OPAS Method

The OPAS method (osmotic pressure for the activity of the solvent) was implemented in *ms*2. This method is an alternative to standard methods, e.g. the Widom test particle method [11] or thermodynamic integration [30, 31, 32], for calculating chemical potentials in the liquid phase in MD simulations. It is particularly well-suited for studying the solvent activity in electrolyte solutions, but can in principle also be used for studying mixtures of molecular species. Details and a thorough assessment of the method are presented in ref. [33], the method is only briefly introduced in the following.

The basic idea is a direct simulation of the osmotic equilibrium between a pure solvent phase and a solution phase by introducing semipermeable membranes in the simulation box. These are realized by an external force field that acts only on the solute molecules to keep them in the solution phase. By sampling the total net membrane force per membrane area, the pressure difference between the two phases, called the osmotic pressure Π, is monitored. Assuming an incompressible solvent, the solvent activity is related to this osmotic pressure by

$$\ln a_{\mathrm{solv}} = -\frac{\Pi v_{\mathrm{solv}}}{RT}, \tag{14}$$

where v_{solv} is the molar volume of the pure solvent at the temperature T, which is easily available from separate, standard molecular simulations. If an electrolyte solution is considered, the activity coefficient of the salt can be obtained by performing OPAS simulations at various salt molalities and applying the Gibbs-Duhem equation to the results for the solvent activity.

Fig. 5 shows molecular simulation results for the mean ionic activity coefficient of NaCl in aqueous solution at T = 298.15 K. There, results obtained by different groups employing different computational approaches are presented. All groups studied the molecular models by Joung and Cheatham [34] for Na^+ and Cl^- ions for use together with SPC/E water, and all simulation results are in mutual agreement.

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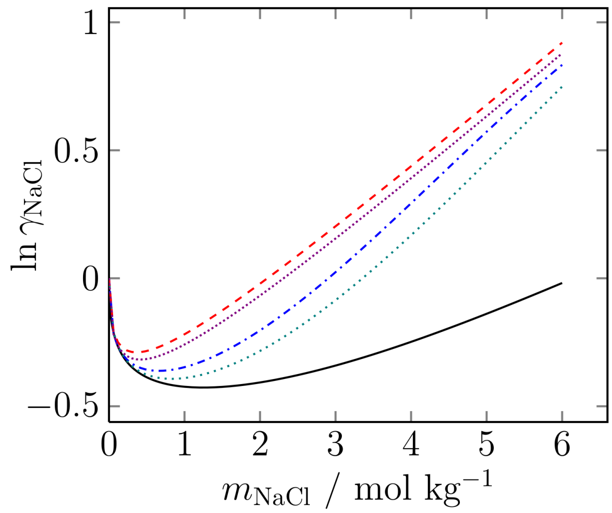

Figure 5: Mean ionic activity coefficient of NaCl over the salt molality for aqueous NaCl solutions at $T = 298.15$ K using the SPC/E + Joung-Cheatham [34] model combination. The colored lines represent simulation results by different groups for that model combination (from top to bottom): OPAS simulations by Kohns *et al.* [33] (red dashed line), free energy calculations by Benavides *et al.* [35] (violet densely-dotted line), gradual insertion of ion pairs by Mester and Panagiotopoulos [31] (blue dashed-dotted line), and osmotic ensemble Monte Carlo simulations by Moučka *et al.* [32] (green dotted line). The black solid line shows the correlation to experimental data by Hamer and Wu [36].

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